

Production data systems and the data engineering behind research: Airflow, DuckDB and PostGIS pipelines that turn open spatial data into papers. Leads a team of 4 as Head of Data; interest in research on urban dynamics.

TOOLS

Python · SQL · PostGIS · Airflow · Iceberg · DuckDB · Docker · PyTorch · Bash

PROFESSIONAL EXPERIENCE

Head of Data, KR&A, Amsterdam

Jul 2025 – present

- Lead a team of 4 through a transformation of the product offering, serving pension funds, real estate investors and national statistical offices.
- Develop, maintain and expand data pipelines (e.g. a weekly scrape of 300k records) and spatial big data products.
- Delivered a global connectivity score: 1TB+ processed across 13 servers into a production API.
- Represent the data function with clients: defending methodology against PhD-level scrutiny, presenting to portfolio managers and senior stakeholders.
- Drive AI adoption: OCR and LLM document extraction, agentic pipeline monitoring, agent-assisted development.

Independent Researcher, urban dynamics

2025 – present

- Self-directed research programme: two working papers and a method paper in preparation (see Research), each backed by an open, reproducible pipeline.

Medior Data Scientist, KR&A

Jun 2022 – Jul 2025

- Project lead for two multi-year projects, including a hedonic house-price-index study for Eurostat, published as an official Eurostat statistical working paper: scraping, storing and managing the data, building the regressions, interpreting results.
- Restructured data infrastructure from legacy systems to Airflow, Iceberg and FastAPI.
- Client-facing throughout, with CBS, Eurostat and pension funds.

Junior Data Scientist, KR&A

Oct 2021 – Jun 2022

- Flood-occurrence prediction from alternative data (BSc-thesis project); 90%+ accuracy in risk classification.
- Improved API efficiency tenfold through spatial optimisations, resulting in promotion.

Junior Full-Stack Developer, Exact (former SRXP), part-time

Sep 2019 – Sep 2022

- Enterprise expense-declaration software (EmberJS, PHP) under CI/CD and testing.
- Maintained a client-facing webapp, working on business logic and styling.

RESEARCH TOPICS

- **Calibrating Free Postcode Boundaries from OpenStreetMap**, Release August 2026. Seed-density-to-IoU calibration of an OSM-Voronoi pipeline; NL/DK references, transfer to BE, applied to Italy's 4,209 CAP polygons.
- **Building an Agent-Based Model to Explain Gentrification in European Cities** (MSc thesis, 8/8): Amsterdam · Utrecht · Milan.

EDUCATION

MSc Data Science & Business Analytics, Bocconi University · 107/110

2023 – 2025

Focus: Finance · Econometrics · Statistics · NLP

Thesis: Building an Agent-Based Model to Explain Gentrification in European Cities

BSc Artificial Intelligence, University of Amsterdam · 7.6/10

2019 – 2023

Minor in Linguistics, University of Amsterdam

Erasmus minor, Università di Bologna (UNIBO)

LANGUAGES & SOFT SKILLS

Dutch C2 · English C2 · Italian B2/C1 · German B2 · Spanish A2

Multi-year project management · Client & stakeholder communication · Presenting and defending methodology to technical and senior audiences · Dependable

PROJECTS

Research pipelines as production systems

Airflow 3 pipelines hosting every personal research project, scaling from a laptop to a multi-machine CeleryExecutor cluster; idempotency guards, custom operators and agentic monitoring.

[Airflow](#) · [Celery](#) · [Docker](#) · [CI](#) · [DuckDB](#)

LanguageBuddy: AI language tutor

Self-hosted AI language tutor for Dutch, Italian and Spanish: chat or voice-call LLM tutor with SM-2 spaced repetition; every mistake feeds the next day's exercises, with a real-news reader and 6,200+ CEFR-aligned vocabulary entries.

[FastAPI](#) · [LLM](#) · [TTS](#) · [SQLite](#) · [Docker](#)

FishFinder: photo-to-species ID

Flutter app identifying 63 Dutch fish species from a photo, fully on-device: ~3,000 hand-annotated photos masked with Segment Anything, a fine-tuned ResNet50 compressed to an 8.8 MB TFLite model.

[Flutter/Dart](#) · [TFLite](#) · [ResNet50](#) · [Segment Anything Model \(SAM\)](#) · [Firebase](#)